

Problem (1)

The three membranes have different concentration profiles,

for membrane a)

$$\cancel{\theta} = \frac{C_m}{C_L} < 1$$

$$\cancel{\theta} = \frac{C_m}{C_R} < 1$$

for membrane (b)

$$\cancel{\theta} = \frac{C_m}{C_L} > 1$$

$$\cancel{\theta} = \frac{C_m}{C_R} < 1$$

for membrane (c)

$$\cancel{\theta} = \frac{C_m}{C_L} < 1$$

$$\cancel{\theta} = \frac{C_m}{C_R} > 1$$



Problem (2)

$$\cancel{\frac{dc}{dt}} = D \frac{1}{r} \frac{d}{dr} \left(r \frac{dc}{dr} \right) = 0$$

$$\frac{dc}{dr} = \frac{A}{r}$$

$$c = B + A \ln(r)$$

$$\phi_1 = \phi_2 \longrightarrow \begin{aligned} c_1(R_i) &= c_i \\ c_2(R_o) &= c_o \\ c_1(R_i) &= c_2(R_o) \end{aligned}$$

$$-D_1 \left. \frac{dc_1}{dr} \right|_{R_i} = -D_2 \left. \frac{dc_2}{dr} \right|_{R_o}$$

$$B_1 = c_i - A_1 \ln R_i$$

$$B_2 = c_o - A_2 \ln(R_o)$$

$$c_i + A_1 \ln(R_i/R_i) = c_o + A_2 \ln(R_i/R_o)$$

$$A_1 = \frac{c_i - c_o + A_2 \ln(R_i/R_o)}{\ln R_i/R_i}$$

$$\left(\frac{C_o - C_i + A_2 \ln(R_1/R_o)}{\ln(R_1/R_i)} \right) \frac{D_1}{R_1} = \frac{A_2 D_2}{R_1}$$

$$A_2 = \frac{D_1 (C_o - C_i)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_o)}$$

$$A_1 = \frac{D_2 (C_o - C_i)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_o)}$$

so

$$C_1(r) = C_i + \left(\frac{D_2 (C_o - C_i)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_o)} \right) \ln(r/R_i)$$

$$C_2(r) = C_o + \left(\frac{D_1 (C_o - C_i)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_o)} \right) \ln(r/R_o)$$